PROJECT REPORT

Customer Shopping Behavior and Trends Analysis

Submitted by

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1. Introduction

Retail businesses generate large volumes of transactional data every day, yet much of the value in that data lies hidden until it is systematically explored. This project applies Exploratory Data Analysis (EDA) and data visualization techniques to a real-world retail dataset in order to uncover purchasing patterns, demographic trends, and the influence of promotional discounts on customer behavior. The insights generated are intended to help a retail business optimize inventory planning, refine marketing campaigns, and improve customer segmentation.

2. Project Objectives

- Clean and preprocess the raw dataset, handling missing values, duplicates, and inconsistent data types.
- Perform EDA to analyze spending distributions across age groups, gender, and product categories.
- Build a visualization dashboard (bar charts, distribution plots, and category breakdowns) to communicate insights clearly.
- Quantify the impact of discounts and promo codes on purchase amount using statistical testing.
- Translate findings into actionable business recommendations.

3. Dataset Description

The analysis uses the "Shopping Trends and Customer Behaviour Dataset," containing 3,900 customer transaction records across 18 attributes, including demographics (age, gender), purchase details (item, category, amount, location, season), and behavioral factors (payment method, shipping type, purchase frequency, and discount/promo usage). The dataset was found to be complete and well-structured, with no missing values and no duplicate records, requiring only minor standardization of categorical fields before analysis.

Attribute	Value
Records	3,900 transactions
Attributes	18 columns
Missing values	None
Duplicate rows	None
Average purchase amount	$59.76
Average customer age	44.1 years
Average previous purchases	25.4

4. Methodology

The analysis was carried out in Python using pandas and numpy for data manipulation, and matplotlib/seaborn for visualization, structured into the following stages:

- Data Cleaning: verifying data types, checking for missing values and duplicates, and standardizing categorical text fields.
- Univariate & Bivariate EDA: examining distributions of age, spend, and ratings, and relationships between spend and demographic/behavioral variables.
- Visualization: category, seasonal, and demographic breakdowns rendered as bar charts, line plots, and pie charts.
- Statistical Testing: an independent-samples t-test comparing purchase amounts between discounted and full-price transactions.

5. Key Findings

5.1 Customer Base Overview

The customer base skews male, with men accounting for 68% of transactions (2,652 of 3,900) against 32% for women. The average purchase amount across all transactions is $59.76, with a fairly tight spread (standard deviation ≈ $23.7), and the average customer has made about 25 previous purchases, indicating a largely repeat-purchase customer base.

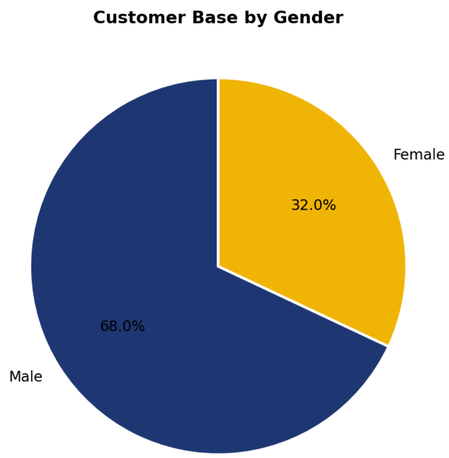

Figure 1. Customer base split by gender.

5.2 Spend by Age Group

Average spend is remarkably stable across age groups, ranging narrowly between $58.88 and $60.65. The 18–25 and 46–55 age bands show the highest average spend (~$60.6 each), while customers aged 66+ spend marginally less on average ($58.88). This suggests age is not a strong driver of transaction size in this dataset, and marketing spend need not be heavily age-segmented on this basis alone.

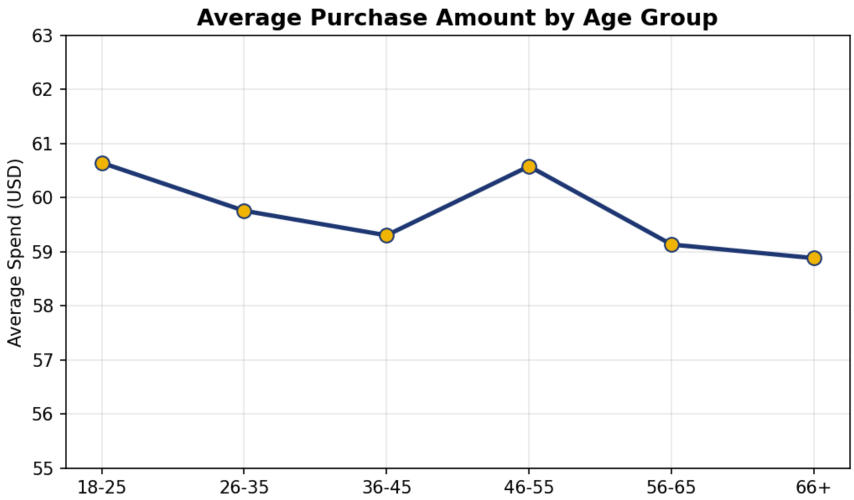

Figure 2. Average purchase amount by age group.

5.3 Revenue by Product Category

Clothing is the dominant category, generating $104,264 in total revenue from 1,737 purchases, followed by Accessories ($74,200 from 1,240 purchases), Footwear ($36,093), and Outerwear ($18,524). Interestingly, average spend per transaction is broadly similar across categories (roughly $57–$60), meaning Clothing's revenue lead comes from purchase volume rather than higher-value items.

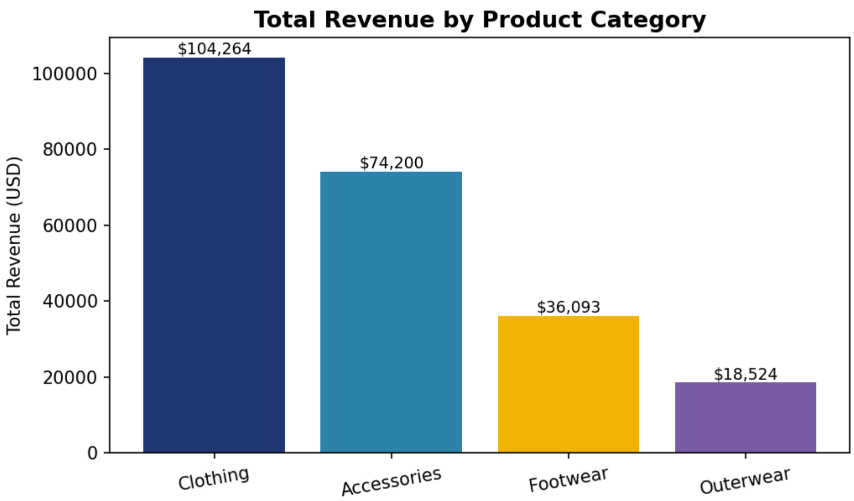

Figure 3. Total revenue generated by each product category.

5.4 Seasonal Trends

Purchase volume and revenue are fairly evenly distributed across the four seasons (955–999 purchases each), with Fall generating the highest total revenue ($60,018) and the highest average spend per transaction ($61.56), followed closely by Winter. Summer is the softest season for both revenue and average spend, suggesting Fall/Winter as the priority window for inventory build-up and promotional pushes.

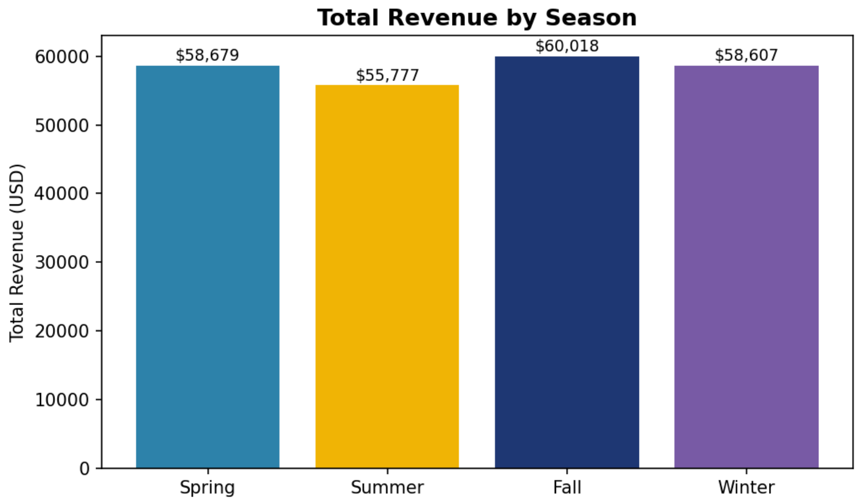

Figure 4. Total revenue by season.

6. Discount & Promo Code Impact Analysis

Discounts and promo codes were used together in 43% of transactions (1,677 of 3,900), and usage is heavily concentrated among subscribed customers — 100% of subscribed customers' purchases involved a discount, compared to just 21.9% for non-subscribers. This points to discounting operating largely as a subscription-loyalty mechanic rather than an open promotional lever.

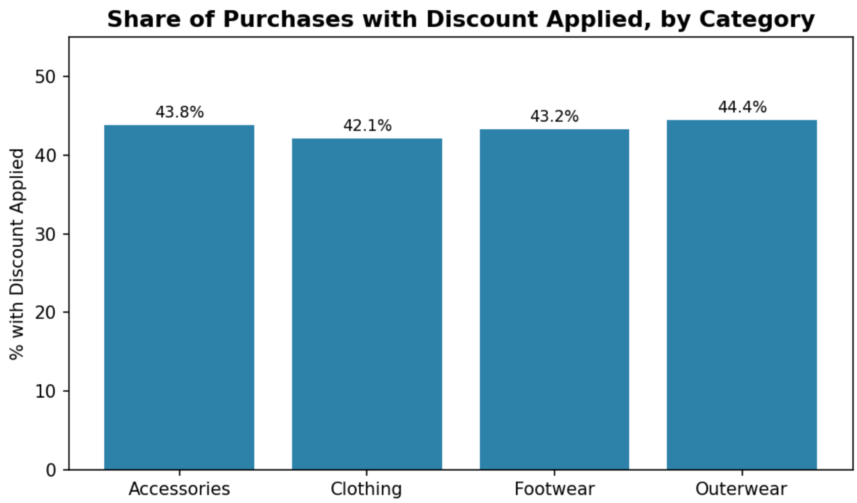

Figure 5. Share of purchases with a discount applied, by category.

To test whether discounting actually changes basket size, an independent-samples t-test was run comparing purchase amounts for discounted ($59.28 average) versus full-price ($60.13 average) transactions. The result ($t = -1.112$, $p = 0.2661$) shows no statistically significant difference between the two groups ($p \geq 0.05$). In other words, discounts in this dataset are not associated with customers spending more or less per transaction — their primary role appears to be rewarding loyalty rather than driving up basket value.

7. Tools & Code Overview

This section gives a plain-language walkthrough of the Python packages used and the key pieces of code behind the analysis, for readers who want to understand how the results in Section 5 and 6 were produced.

7.1 Packages Used

- pandas — the core library for loading the CSV, cleaning it, and slicing/grouping data (e.g., computing averages per category or age group).
- numpy — used underneath pandas for fast numerical operations such as means, standard deviations, and array handling.
- matplotlib — the base plotting library used to draw and format all charts (bar charts, line plots, pie charts).
- seaborn — built on matplotlib; used to apply a cleaner visual style and for some distribution-style plots.
- scipy.stats — provides the t-test (ttest_ind) used to check whether discounted and full-price purchases differ significantly in amount.

7.2 Key Code Explained

Loading and inspecting the data:

```
df = pd.read_csv('Shopping_Trends_And_Customer_Behaviour_Dataset.csv')
print(f"Dataset shape: {df.shape[0]} rows x {df.shape[1]} columns")
```

This reads the raw CSV into a pandas DataFrame (a table-like structure) and prints its size, confirming 3,900 rows and 18 columns before any cleaning.

Checking data quality:

```
df.isnull().sum()          # counts missing values per column
df.duplicated().sum()      # counts fully duplicated rows
```

isnull().sum() flags any empty cells column by column, and duplicated().sum() counts rows that are exact repeats of another row. Both returned zero, meaning no cleaning of missing or duplicate data was required.

Creating an age group column:

```
df['Age Group'] = pd.cut(
    df['Age'],
    bins=[17, 25, 35, 45, 55, 65, 100],
    labels=['18-25','26-35','36-45','46-55','56-65','66+']
)
```

pd.cut() converts the continuous Age column into labeled buckets (e.g., “18-25”), which makes it possible to compare average spend across age bands rather than individual ages.

Grouping and aggregating (the core of the EDA):

```
category_summary = df.groupby('Category').agg(
    Purchases=('Category', 'count'),
    Total_Revenue=('Purchase Amount (USD)', 'sum'),
    Avg_Spend=('Purchase Amount (USD)', 'mean')
)
```

groupby('Category') splits the data into one group per product category, and .agg() then computes the purchase count, total revenue, and average spend within each group in a single step. The same pattern (groupby + agg) is reused for Age Group, Season, and Gender breakdowns throughout Section 5.

Cross-tabulating two categorical columns:

```
discount_by_subscription = pd.crosstab(
    df['Subscription Status'], df['Discount Applied'],
    normalize='index'
) * 100
```

pd.crosstab() builds a percentage table showing, for each Subscription Status (Yes/No), what share of purchases had a discount applied. normalize='index' converts raw counts into row-wise percentages, which is what produced the 100% vs 21.9% discount-usage figures in Section 6.

Testing statistical significance:

```
from scipy import stats
discounted = df[df['Discount Applied Flag'] == 1]['Purchase Amount (USD)']
not_discounted = df[df['Discount Applied Flag'] == 0]['Purchase Amount (USD)']
t_stat, p_value = stats.ttest_ind(discounted, not_discounted, equal_var=False)
```

This splits transactions into two groups (discounted vs. not) and runs an independent-samples t-test to check if their average purchase amounts differ by more than chance would explain. A p-value below 0.05 would indicate a real difference; here $p = 0.2661$, so the difference is not statistically significant.

Plotting a chart (the same pattern used for every chart in this report):

```
plt.figure(figsize=(9, 5))
category_summary['Total_Revenue'].sort_values(ascending=False).plot(kind='bar')
plt.title('Total Revenue by Product Category')
plt.ylabel('Total Revenue (USD)')
plt.tight_layout()
plt.show()
```

plt.figure() sets the canvas size, .plot(kind='bar') draws the bar chart directly from the aggregated data, and the title/label calls make it readable. This same three-step pattern (aggregate → plot → label) underlies every visualization in Section 5.

8. Business Recommendations

- Prioritize Clothing and Accessories in inventory and marketing spend, as they jointly account for over 76% of category revenue.
- Time major promotional campaigns and stock replenishment around Fall and Winter, the two highest-revenue seasons.
- Re-evaluate blanket discounting: since discounts show no significant effect on transaction value, position them explicitly as a subscription/loyalty benefit rather than a broad sales lever.
- Since spend is fairly flat across age groups, focus segmentation efforts on behavioral variables (subscription status, purchase frequency, category preference) rather than age alone.
- Track payment method and shipping type distributions periodically to ensure operational capacity (e.g., express shipping, digital wallets) matches evolving customer preference.

9. Conclusion

This EDA of 3,900 retail transactions shows a stable, repeat-purchase customer base with fairly uniform spending across age groups, a clear revenue concentration in Clothing and Accessories, and modest seasonal peaks in Fall and Winter. Most notably, discount and promo code usage functions primarily as a subscription-loyalty signal rather than a driver of higher spend, since statistical testing found no significant difference in purchase amount between discounted and full-price transactions. These findings give the business a data-backed basis for prioritizing inventory categories, timing campaigns seasonally, and rethinking the strategic purpose of its discounting program.